Imen Al Samarai

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Last updated October 2026

EXPERIENCE

Founder & CEO

[Year] – present

Pontus Analytics · Paris

- AI, machine learning and physics-simulation solutions for quantitative finance, defence and R&D.
- Research on evaluating how language-model agents make decisions under uncertainty.

Director, Quantitative Research

Oct 2020 – Jun 2024

TD Securities · London

- Built the systematic fixed-income research platform from concept to firm-wide adoption across trading desks.
- Grew the team from individual contributor to seven PhD-level researchers, extending research to FX and commodities.
- Developed machine-learning alpha signals with validated backtesting, and NLP tools for central-bank sentiment.

Lead Data Scientist & ML Engineer

Nov 2018 – Oct 2020

Swiss Life · Paris

- Designed and deployed a production robo-advisory engine combining LSTM models and genetic algorithms, serving thousands of clients.
- Built the full ML pipeline — client profiling, rebalancing, real-time risk monitoring — and led a team of three data scientists.

Senior Research Scientist

Aug 2016 – Nov 2018

CERN · Geneva

- Machine-learning methods in the search for the cosmic sources of neutrinos, contributing to results published in Science.
- Coordinated work across international collaborations and supervised three PhD researchers.

Researcher

May 2013 – Aug 2016

LPNHE · Paris

- Developed a phenomenological model and a C++ simulation framework adopted across four research projects.
- Co-led competitive technical proposals for experimental deployment.

Visiting Scientist · Assistant Professor

Jan 2012 – Apr 2013

University of Michigan · Aix-Marseille University · Ann Arbor · Marseille

- Built real-time image-processing pipelines for autonomous telescope networks detecting transient events.
- Taught physics, mathematics and statistics.

EDUCATION

PhD, Particle Physics & Astrophysics

2009 – 2012

Aix-Marseille University · Signal detection in high-noise astrophysical data

MSc, Particle Physics

2008

Université Paris 7 Denis Diderot

PUBLICATIONS

200+ publications in astroparticle physics and AI, including Science and Nature.

SKILLS

Methods

Statistical inference, Monte Carlo simulation, signal processing, econometrics, portfolio optimisation

Machine learning

Classical ML, deep learning, reinforcement learning, LLMs and agentic systems; PyTorch, TensorFlow, scikit-learn

Engineering

Python, C++, SQL, R; Docker, AWS, Spark, Airflow; Bloomberg and Reuters APIs

RECOGNITION & LANGUAGES

Legendary Award, TD Securities	2023
Above and Beyond Award, TD Securities	2021
NSF press release on the Science neutrino result	2018

French, Arabic (native) · English (fluent) · Italian, Spanish (conversational)